Research Article

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Effects of dietary polysaccharides from the submerged fermentation concentrate of *Hericium caput-medusae* (Bull.:Fr.) Pers. on fat deposition in broilers

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Abstract

BACKGROUND: The present study was conducted to investigate the lipid-lowering effect of polysaccharides from the submerged fermentation concentrate of *Hericium caput-medusae* (Bull.:Fr.) Pers. (HFCP) in broilers. A total of 480 female Arbor Acres broilers were randomly divided into four dietary treatments, each consisting of six pens as replicates, and fed diets containing 0 (control), 1, 3 or 5 g kg^{-1} HFCP.

RESULTS: The results revealed that the average daily gain of broilers increased (linear (L), $P < 0.01$; quadratic (Q), $P < 0.01$) when the HFCP levels increased. The serum cholesterol, triglyceride and low-density lipoprotein cholesterol levels decreased (Q, $P < 0.05$) while the high-density lipoprotein cholesterol level increased (Q, $P < 0.05$) when the HFCP levels increased. The caecum *Escherichia coli* count and pH decreased (Q, $P < 0.01$) while the lactobacilli count and bifidobacteria count increased (L, $P < 0.05$; Q, $P < 0.05$) when the HFCP levels increased. The propionic acid and butyric acid concentrations increased (L, $P < 0.001$; Q, $P < 0.001$) while the abdominal fat rate and liver fat content decreased (L, $P < 0.01$; Q, $P < 0.05$) when the HFCP levels increased.

CONCLUSION: Dietary supplementation with HFCP may lead to the development of low abdominal fat of broilers as demanded by health-conscious consumers.
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Keywords: *Hericium caput-medusae* (Bull.:Fr.) Pers; polysaccharides; broilers; abdominal fat; gut microflora; SCFA

INTRODUCTION

For several decades the commercial broiler industry has focused on enhancing bird growth rate and feed conversion efficiency.[1] However, this breeding strategy has led to inevitable problems such as a decline in meat quality and a concomitant increase in body fat deposition.[2] The relatively larger amount of body fat in animal products, which is a risk factor for cardiovascular diseases, is not welcomed by consumers.[3] Furthermore, the discarded abdominal and visceral fat increases waste management problems.[4] In recent years, there has been increased awareness of the benefits of functional food additives.

For centuries, edible mushrooms have been used to maintain health and increase longevity; they were first used by ancient civilizations.[5] Mushrooms contain substances, mainly polysaccharides, polysaccharopeptides and polysaccharide proteins, with antitumor, antiviral and antibacterial properties.[6,7] The hypolipidemic effects of polysaccharides from a few mushrooms have already been studied. It has been reported that chitosan extracted from *Agaricus bisporus*[6] and the exo-biopolymer (total sugar content 912 g kg^{-1}) from a submerged mycelial culture of *Hericium erinaceus*[8] reduced the serum triglyceride concentration in dietary-induced hyperlipidemic rats.

The polysaccharides of mushrooms can be obtained from fruit bodies, mycelia and culture filtrates.[9,10] However, when macro-fungi grow on solid culture, the growth cycle to complete the fruiting body is too long and product quality control is difficult. Submerged culture is an alternative way to produce large quantities of macro-fungal polysaccharides. There are several advantages of submerged culture over solid culture for polysaccharide production: shorter fermentation period, lower cost, more convenient control, higher product concentration and easier downstream processing.[11–13]

Hericium caput-medusae (Bull.:Fr.) Pers. is a well-known edible and medicinal mushroom in oriental countries. However, there

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have been few studies on the physiological effects of polysaccharides from *H. caput-medusae* (Bull.:Fr.) Pers., particularly on lipid metabolism. It was hypothesized that dietary polysaccharides might affect lipid metabolism by acting on gut microflora and caecum short-chain fatty acids (SCFA).[14] The present study was designed to investigate the effects of dietary supplementation with polysaccharides from the submerged fermentation concentrate of *H. caput-medusae* (Bull.:Fr) Pers. (HFCP) on abdominal fat deposition, serum lipid profiles, intramuscular fat content, gut microflora and pH, caecum SCFA as well as fecal fat excretion of broilers, which would facilitate its use as a possible additive to produce broilers with lower fat deposition.

MATERIALS AND METHODS

Preparation of HFCP

The submerged fermentation concentrate of *H. caput-medusae* (Bull.:Fr.) Pers. (HFC) was supplied by the Bethune Pharmaceutical Factory of Jilin University, China. Polysaccharides from HFC were prepared as follows. To remove lipids, HFC was treated with 950 g L^{-1} ethanol at room temperature for 24 h with intermittent stirring, and the extraction was repeated three times. The residues were extracted with hot water (80 °C, 2 volumes) for 4.5 h. The extract was precipitated with 3 volumes of 950 g L^{-1} ethanol and centrifuged. The precipitate was dried using vacuum freeze-drying equipment (FD-1B, Beijing Boyikang Experimental Instrument Co., Ltd, Beijing, China) to yield the water-extractable polysaccharides that were added to the chicken feed. The content of polysaccharides in HFCP was determined by the phenol/sulfuric acid method.[15] Calcium, phosphorus and selenium contents and the proximate composition (including dry matter, crude protein, ether extract, crude ash and crude fiber) of HFCP were analyzed according to AOAC methods.[16] The ingredients of HFCP are shown in Table 1.

Birds, diets and management

A total of 480 1-day-old female Arbor Acres broilers were purchased from Ronda Animal Husbandry Company (Yitong, Jilin, China). The broilers were randomly allotted to four groups in six replicates, with each replicate comprising 20 broilers per pen. All feeds were formulated to meet the nutrient requirements for broilers suggested by the NRC.[17] Feeds were supplemented with 0 (control), 1, 3 or 5 g kg^{-1} HFCP in powder form. The experiment was conducted in two phases: a starter period from 1 to 21 days old and a finisher period from 22 to 42 days old. The proximate composition of the feeds was analyzed according to AOAC methods.[16] The composition and nutrient contents of the feeds are shown in Table 1. The feed mixtures contained no anticoccidial or antibacterial supplements. Feeds and water were provided *ad libitum*.

The feeding experiment was carried out in Jilin Agricultural University, with all procedures approved by the Animal Care and Use Committee of Jilin Agricultural University, according to the Regulations for the Administration of Affairs Concerning Experimental Animals (approved by the State Council of the People's Republic of China on 31 October 1988 and promulgated by Decree no. 2 of the State Science and Technology Commission of the People's Republic of China on 14 November 1988). All chickens were placed in a temperature-controlled house. The temperature was maintained at 33 ± 1 °C during the first 3 days, after which it was gradually lowered by 3 °C a week until reaching 24 °C. The temperature of the room was then maintained at 24 °C for the remainder of the study. Light was provided 24 h per day using fluorescent lights.

Sample collection

On days 21 and 42 the chickens were fasted for 12 h. The chickens and feeds were then weighed to determine the average daily gain (ADG), the average daily feed intake (ADFI) and the feed conversion (ADFI/ADG).

On days 21 and 42, 24 birds were randomly selected from each group (four birds per pen) and blood samples were withdrawn from the wing vein into a sterile syringe. Samples for serum analysis were then centrifuged at 3000 × *g* for 15 min and serum was separated. The serum samples were stored at −20 °C until analyzed for total cholesterol (TC), triglyceride (TG), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), alkaline phosphatase (ALP), aspartate aminotransferase (AST) and alanine aminotransferase (ALT).

Following blood collection, the same birds were weighed individually and euthanized by cervical dislocation. The abdominal fat of each bird was removed and weighed. The digesta of the right caecum were collected for the measurement of pH and bacterial counts immediately. The digesta of the left caecum were removed and stored at −80 °C for the analysis of SCFA. The left liver lobe and left thigh (*biceps femoris*) and breast (*pectoralis major*) muscles without skin and adipose tissues were collected from the same location of each chicken, washed with saline, blotted dry on filter paper and chopped. Excreta from day 19 to day 21 and from day 40 to day 42 were collected from each pen and pooled within pen. Liver, thigh, breast and excreta samples were dried at 60 °C, ground to pass through a 40-mesh screen and mixed thoroughly before the analysis of fat content.

Determination of serum indices

The concentrations of TC, TG, HDL-C, LDL-C, ALP, AST and ALT in the serum samples were analyzed with an automatic biochemical analyzer (WD-98100525, Shenzhen Mindray Bio-Medical Electronics Co., Ltd, Shenzhen, China) using colorimetric methods, following the instructions of the manufacturer of the corresponding reagent kit (Zhongsheng Biochemical Co., Ltd, Beijing, China).

Gut microflora examinations and pH determination

Approximately 1 g of each digesta sample was mixed with 9 mL of sterile NaCl solution (9 g L^{-1}) and homogenized for 3 min. From the initial 10^{-1} dilution, tenfold serial dilutions were subsequently made with the same diluent. Dilutions of the samples were then inoculated in duplicate onto selective agars (Qingdao Haibo Biological Technology Co., Ltd, Qingdao, China): MacConkey agar (HB7001) for *Escherichia coli*, lactobacilli anaerobic MRS agar (HB0384) for lactobacilli and bifidobacteria BS agar (HB0394) for bifidobacteria. Inoculated agar plates were incubated in an anaerobic environment at 37 °C for 48 h, except for MacConkey agar plates, which were incubated aerobically at 37 °C for 24 h. The colonies on each plate were counted as soon as the plates were removed from the incubator. Microbial counts were expressed as $\log_{10}$ colony-forming units (CFU) g^{-1} wet digesta for each digesta sample.[18]

The pH of the caecum digesta was determined using a pH meter (PHS-3C, Shanghai Pengshun Scientific Instrument Co., Ltd, Shanghai, China). A 1 g aliquot of each digesta sample was diluted with 9 mL of distilled water for pH measurement.

Analysis of SCFA in caecum digesta

The SCFA content in 2 g of each caecum digesta sample was extracted in 5 mL of water by vortexing for 5 min in a cold room.

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Table 1. Ingredients and chemical composition of experimental diets and HFCP

	Starter period (days 1–21)				Finisher period (days 22–42)			
	Levels of dietary HFCP (g kg^{-1})				Levels of dietary HFCP (g kg^{-1})			
Item	0	1	3	5	0	1	3	5
Ingredients (g kg^{-1})								
Corn	561.0	560.4	559.2	558.0	595.0	594.3	593.1	591.8
Soybean meal (44%)	359.0	358.6	357.8	357.0	330.0	329.7	328.9	328.2
Corn oil	35.0	35.0	35.0	35.0	35.0	35.0	35.0	35.0
$CaHPO_4 \cdot 2H_2O$	19.0	19.0	19.0	19.0	13.0	13.0	13.0	13.0
Limestone	11.0	11.0	11.0	11.0	13.0	13.0	13.0	13.0
HFCP powder	0.0	1.0	3.0	5.0	0.0	1.0	3.0	5.0
Salt	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
DL-Methionine	2.0	2.0	2.0	2.0	1.0	1.0	1.0	1.0
Premix[a]	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Total	1000.0	1000.0	1000.0	1000.0	1000.0	1000.0	1000.0	1000.0
Chemical analysis (g kg^{-1})								
Crude protein	220.9	220.8	220.9	220.7	209.9	209.8	209.8	209.9
Crude fiber	21.9	21.8	21.9	21.9	23.2	23.3	23.2	23.3
Ether extract	53.4	53.3	53.5	53.3	55.2	55.3	55.2	55.2
Crude ash	60.9	61.0	60.8	60.8	62.3	62.5	62.4	62.4
Calcium	9.6	9.6	9.7	9.6	8.8	8.9	8.9	8.8
Phosphorus	4.6	4.7	4.7	4.6	3.5	3.6	3.6	3.6
Calculated values								
ME (MJ · kg^{-1})	12.9	12.9	12.9	12.9	13.3	13.3	13.3	13.3
Lysine (%)	1.30	1.30	1.30	1.30	1.10	1.10	1.10	1.10
Threonine (%)	0.85	0.85	0.85	0.85	0.78	0.78	0.78	0.78
TSAA (%)	0.89	0.89	0.89	0.89	0.83	0.83	0.83	0.83
HFCP composition (g kg^{-1})[b]								
Dry matter, 944.6 ± 2.8; crude protein, 58.2 ± 0.7; ether extract, 10.2 ± 0.3; crude ash, 45.9 ± 0.5; crude fiber, 1.6 ± 0.1; nitrogen-free-extract, 828.7; polysaccharide content, 774.7 ± 2.13; phosphorus, 2.2 ± 0.1; calcium, 2.6 ± 0.1; selenium, 0.073 ± 0.004								

HFCP, polysaccharides from submerged fermentation extract of *Hericium caput-medusae* (Bull.:Fr.) Pers.; ME, metabolizable energy; TSAA, total sulfur amino acids.
[a] Supplied (kg^{-1} diet): vitamin A (retinyl palmitate), 9000 IU; vitamin D_3, 2000 IU; vitamin E (DL-α-tocopheryl acetate), 10 mg; vitamin K (menadione), 0.5 mg; vitamin B_1, 1.8 mg; pyridoxine, 3.5 mg; vitamin B_{12} (cyanocobalamin), 0.01 mg; riboflavin, 3.6 mg; niacin, 35 mg; pantothenic acid, 10 mg; folic acid, 0.55 mg; choline chloride, 250 mg; biotin, 0.15 mg; Mn (as $MnSO_4 \cdot H_2O$), 60 mg; Zn (as ZnO), 40 mg; Fe (as $FeSO_4 \cdot H_2O$), 80 mg; Cu (as $CuSO_4 \cdot 5H_2O$), 8 mg; I (as CaI_2), 0.35 mg; Se (as Na_2SeO_3), 0.15 mg.
[b] Values are expressed as mean ± standard deviation ($n = 6$).

The mixture was then centrifuged at 5000 × *g* for 10 min. A 200 μL aliquot of 250 g L^{-1} metaphosphoric acid was added to 1 mL of the supernatant, kept in an ice bath for 30 min and then centrifuged at 10 000 × *g* for 10 min. The supernatant was used to determination the SCFA in digesta. The measurement was carried out using a gas chromatograph (7890A, Agilent, Santa Clara, CA, USA) equipped with a flame ionization detector and a DB-FFAP capillary column (30 m × 0.25 mm, 0.25 μm). The following oven temperature program was used: initial temperature 65 °C, increasing to 190 °C at 20 °C min^{-1}. A 1 μL sample was injected at a split ratio of 1:50 with an inlet temperature of 220 °C. Commercially available acetic acid, propionic acid and butyric acid (Sigma, Santa Clara, CA, USA) were used as external standards.

Body fat and excreta fat

The body fat level of birds in this experiment was evaluated using four indicators, abdominal fat rate (AFR), liver fat content, intramuscular fat content of thigh muscle (TIF) and intramuscular fat content of breast muscle (BIF). The fat content in liver, muscles and excreta was determined by extraction with ether in a Soxhlet apparatus.[16] The intramuscular fat and excreta fat percentages were calculated.

Statistical analyses

Pen was used as the experimental unit and one-way analysis of variance was performed using the GLM procedure with SAS software (SAS Institute Inc., Cary, NC, USA) in a completely randomized design. Probability values less than 0.05 were considered significant. When significant differences were determined among treatments means, they were separated by using Tukey's HSD test. In addition, orthogonal polynomial contrasts were used to determine significant linear and quadratic response to increasing dietary HFCP levels. Linear correlation was calculated between two sets of parameters.

RESULTS AND DISCUSSION

Broiler production performance

The effects of HFCP on broiler growth are presented in Table 2. During the starter period (days 1–21) the ADG (linear (L), $P < 0.001$;

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Table 2. Effects of dietary HFCP supplementation on production performance of broilers

Item	HFCP (g kg^{-1}) 0 (control)	1	3	5	SEM	*P* value[a] L	Q
Starter period (days 1–21)							
ADG (g day^{-1} per bird)	34.1c	35.8b	37.8a	38.1a	0.37	<0.001	<0.001
ADFI (g day^{-1} per bird)	52.3c	54.8bc	56.8ab	58.1a	0.55	<0.001	<0.001
Feed conversion	1.53	1.53	1.50	1.53	0.01	0.641	0.583
Finisher period (days 22–42)							
ADG (g day^{-1} per bird)	71.1b	71.4b	70.8b	73.1a	0.24	0.007	<0.001
ADFI (g day^{-1} per bird)	153.3b	152.7b	154.7ab	163.8a	1.47	0.004	0.004
Feed conversion	2.16	2.14	2.18	2.24	0.02	0.036	0.085
Overall experiment (days 1–42)							
ADG (g day^{-1} per bird)	52.6d	53.6c	54.3b	55.6a	0.24	<0.001	<0.001
ADFI (g day^{-1} per bird)	102.8b	103.7b	105.7b	110.9a	0.90	<0.001	<0.001
Feed conversion	1.95	1.93	1.95	1.99	0.01	0.140	0.145

Values are mean of six pens. Means in the same row without the same letter are significantly different ($P < 0.05$). ADG, average daily gain; ADFI, average daily feed intake; SEM, standard error of mean.
[a] L, linear effect of increasing HFCP; Q, quadratic effect of increasing HFCP (0, 1, 3 and 5 g kg^{-1} diet).

quadratic (Q), $P < 0.001$) and ADFI (L, $P < 0.001$; Q, $P < 0.001$) of broilers increased when the HFCP levels increased. Similar trends were observed for the finisher period (days 22–42) and the overall experiment (days 1–42). However, no difference in feed conversion was observed among different groups in each period ($P > 0.05$). Previous studies have shown that the inclusion of *Pleurotus eryngii* stalk residue and *A. bisporus* in broiler diet conferred growth-promoting activity.[5,19] Actually, natural medicinal products originating from mushrooms have been used as feed additives for farm animals in China for centuries and shown many bioactivities such as antimicrobial effects, immune enhancement and stress reduction.[20] The bioactive components of these products are quite complex, but, among them, polysaccharides are considered to be important.[20] It is well documented that polysaccharides derived from *Lentinus edodes* can act as an alternative to antibiotic growth promoters.[20,21]

Serum characteristics

The effects of HFCP supplementation on serum characteristics of broilers are shown in Table 3. On day 21 the serum TC (Q, $P < 0.05$), TG (L, $P < 0.001$; Q, $P < 0.001$) and LDL-C (L, $P < 0.001$; Q, $P < 0.001$) levels decreased while the HDL-C level increased (Q, $P < 0.05$) when the HFCP levels increased. A similar trend was observed for day 42, and serum TC, TG, LDL-C and HDL-C levels trends were quadratic ($P < 0.05$) with increasing HFCP levels. These results showed that dietary supplementation with HFCP was able to lower blood lipids of broilers. The results are consistent with those of an earlier study in which the exo-biopolymer (total sugar content 912 g kg^{-1}) from a submerged mycelial culture of *H. erinaceus* was found to lower the concentration of blood lipids in rats.[8] High-density lipoproteins are a class of lipoproteins that vary somewhat in size (8–11 nm in diameter).[22] These lipoproteins carry fatty acids and cholesterol from the body's tissues to the liver for excretion or reutilization. Dietary supplementation with HFCP in broiler chicken feed increased the serum concentration of HDL-C and decreased that of LDL-C, improving the clearance rate of body cholesterol. Triglycerides are secreted from the liver and carried into the blood by triglyceride-rich lipoproteins; impairment of hepatic lipogenesis results in decreased triglyceride concentrations in plasma. In this study the TG concentration in blood decreased in response to treatment with HFCP, suggesting that dietary supplementation with HFCP might affect hepatic lipogenesis of broilers.

Any abnormal increase in serum levels of ALP, AST and ALT can imply liver damage (hepatocellular degeneration).[23] However, there were no significant alterations in the activity of serum ALP, AST and ALT from broilers with HFCP compared with the control group in this study (Table 3).

Bacterial counts, pH and SCFA in caecum

The effects of HFCP supplementation on bacterial counts and pH in the caecum digesta of broilers are shown in Table 4. On day 21 the caecum *E. coli* count (L, $P < 0.01$; Q, $P < 0.01$) and pH (L, $P < 0.001$; Q, $P < 0.001$) decreased while the lactobacilli count (L, $P < 0.001$; Q, $P < 0.01$) and bifidobacteria count (L, $P < 0.05$; Q, $P < 0.05$) increased when the HFCP levels increased. A similar trend was observed for day 42.

In poultry production, antibiotics have been routinely used to prevent disease and improve feed efficiency. However, public concerns over routine use of antibiotics in livestock feeding have increased owing to the emergence of antibiotic-resistant bacteria that may represent a risk to human health. This threat drove the European Union to ban the use of antibiotics in livestock rearing.[19] Consequently, considerable effort has been directed towards developing alternatives to antibiotics; for example, various oligosaccharides are now being added to livestock feed as prebiotics to improve animal performance and to influence gut microbial flora concentration.[24–26] The intestinal microbial flora of broilers play an important role in growth performance and health of birds.[27] Prebiotics are defined as non-digestible food ingredients that beneficially affect the host by selectively stimulating the growth or activity, or both, of one or more bacteria in the colon.[28,29] HFCP is a polysaccharide that is relatively easily obtained by fungal fermentation. The present study showed that dietary supplementation with HFCP decreased the *E. coli* counts and increased the lactobacilli and bifidobacteria counts in the caecum of birds on both

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Table 3. Effects of dietary HFCP supplementation on serum characteristics of broilers

Item	HFCP (g kg^{-1}) 0 (control)	1	3	5	SEM	*P* value[a] L	Q
Day 21							
TC (mmol L^{-1})	4.55a	3.76b	3.63b	3.87b	0.108	0.056	0.003
TG (mmol L^{-1})	1.24a	0.88b	0.72b	0.75b	0.047	<0.001	<0.001
HDL-C (mmol L^{-1})	2.56b	2.70ab	3.05a	2.84ab	0.065	0.051	0.019
LDL-C (mmol L^{-1})	1.12a	0.76b	0.42c	0.63b	0.057	<0.001	<0.001
ALP (U L^{-1})	366	361	366	377	3.49	0.168	0.248
AST (U L^{-1})	175	173	172	181	1.69	0.226	0.184
ALT (U L^{-1})	16.2	15.5	15.3	17.2	0.33	0.255	0.091
Day 42							
TC (mmol L^{-1})	3.26a	2.74ab	1.51b	2.52ab	0.254	0.186	0.015
TG (mmol L^{-1})	0.78a	0.68ab	0.51c	0.65b	0.038	0.135	0.002
HDL-C (mmol L^{-1})	0.70b	1.29ab	2.03a	1.79a	0.200	0.016	0.001
LDL-C (mmol L^{-1})	0.80a	0.64ab	0.32c	0.53bc	0.067	0.074	0.004
ALP (U L^{-1})	366	364	370	376	2.61	0.089	0.218
AST (U L^{-1})	166	174	173	171	1.53	0.400	0.214
ALT (U L^{-1})	16.8	16.5	16.5	17.2	0.20	0.497	0.411

Values are mean of six pens of four birds each. Means in the same row without the same letter are significantly different ($P < 0.05$). TC, total cholesterol; TG, triglyceride; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; ALP, alkaline phosphatase; AST, aspartate aminotransferase; ALT ,alanine aminotransferase.
[a] L, linear effect of increasing HFCP; Q, quadratic effect of increasing HFCP (0, 1, 3 and 5 g kg^{-1} diet).

Table 4. Effects of dietary HFCP supplementation on bacterial counts ($\log_{10}$ CFU g^{-1} wet digesta) and pH in caecum digesta of broilers

Item	HFCP (g kg^{-1}) 0 (control)	1	3	5	SEM	*P* value[a] L	Q
Day 21							
Escherichia coli	8.69a	7.96b	7.94b	7.73b	0.114	0.002	0.002
Lactobacilli	9.12b	9.35ab	9.49a	9.60a	0.061	<0.001	0.002
Bifidobacteria	9.18b	9.45a	9.42ab	9.54a	0.047	0.014	0.035
pH	6.81a	6.56b	6.41c	6.38c	0.052	<0.001	<0.001
Day 42							
Escherichia coli	7.94a	7.52b	7.39b	7.55b	0.066	0.048	<0.001
Lactobacilli	9.11c	9.44b	9.58a	9.61a	0.061	<0.001	<0.001
Bifidobacteria	9.55b	9.58ab	9.65ab	9.69a	0.020	0.002	0.008
pH	6.91a	6.83a	6.57b	6.41c	0.061	<0.001	<0.001

Values are mean of six pens of four birds each. Means in the same row without the same letter are significantly different ($P < 0.05$).
[a] L, linear effect of increasing HFCP; Q, quadratic effect of increasing HFCP (0, 1, 3 and 5 g kg^{-1} diet).

days 21 and 42, which indicated that HFCP has the properties of a prebiotic.

The effects of dietary HFCP supplementation on caecum SCFA of broilers are shown in Table 5. Dietary supplementation with different levels of HFCP had no effect on acetic acid concentration ($P > 0.05$). On both days 21 and 42 the propionic acid concentration increased (L, $P < 0.001$; Q, $P < 0.001$) when the HFCP levels increased. A similar trend was observed for butyric acid concentration (L, $P < 0.001$; Q, $P < 0.001$).

Yusrizal and Chen[30] indicated that normal intestinal microflora such as *Lactobacillus* spp. or *Bifidobacterium* spp. use inulin or oligofructose (both are prebiotics in common use) for fermentation more efficiently than other groups of bacteria; these microorganisms produce SCFA and lactate on prebiotics to create an acidic environment that suppresses the growth of putrefactive proteolytic bacteria. In the present study, on day 21, lactobacilli counts had a positive linear correlation with propionic acid concentration ($r = 0.989$) and a negative linear correlation with pH ($r = -0.986$) whereas pH had a positive linear correlation with *E. coli* counts ($r = 0.956$) in the caecum of broilers; on day 42, bifidobacteria counts had positive linear correlations with propionic acid concentration ($r = 0.963$) and butyric acid concentration ($r = 0.996$) and a negative linear correlation with pH ($r = -0.998$) in the caecum of broilers, which indicated that the reduction of coliforms in the study may be due to the increased acidity in the caecum.

Fat deposition and fat excretion

The effects of dietary HFCP supplementation on fat deposition and fat excretion of broilers are presented in Table 6. Dietary

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Table 5. Effects of dietary HFCP supplementation on caecum SCFA (μmol g^{-1}) of broilers

Item	HFCP (g kg^{-1})				SEM	*P* value[a]	
	0 (control)	1	3	5		L	Q
Day 21							
Acetic acid	47.7	49.9	50.0	51.8	0.602	0.018	0.067
Propionic acid	2.20c	3.56b	4.15b	5.28a	0.340	<0.001	<0.001
Butyric acid	4.34c	4.69bc	5.22b	6.42a	0.251	<0.001	<0.001
Day 42							
Acetic acid	49.8	50.0	50.9	50.1	0.691	0.806	0.883
Propionic acid	2.03c	2.80b	3.18b	4.00a	0.219	<0.001	<0.001
Butyric acid	2.45b	3.12b	5.09a	5.76a	0.421	<0.001	<0.001

Values are mean of six pens of four birds each. Means in the same row without the same letter are significantly different ($P < 0.05$).
[a] L, linear effect of increasing HFCP; Q, quadratic effect of increasing HFCP (0, 1, 3 and 5 g kg^{-1} diet).

Table 6. Effects of dietary HFCP supplementation on fat deposition and excretion (%) of broilers

Item	HFCP (g kg^{-1})				SEM	*P* value[a]	
	0 (control)	1	3	5		L	Q
Day 21							
AFR	1.11a	0.84ab	0.64b	0.50b	0.080	<0.001	0.003
Liver fat content	11.1a	10.7a	9.60ab	7.70b	0.453	<0.001	0.001
TIF	4.41	4.54	4.34	4.21	0.142	0.519	0.792
BIF	2.46	2.56	2.51	2.41	0.072	0.693	0.795
Excreta fat content	2.38	2.55	2.80	3.22	0.145	0.021	0.079
Day 42							
AFR	2.25a	2.12ab	2.01bc	1.88c	0.052	0.003	0.013
Liver fat content	12.6a	12.6a	11.8ab	11.2b	0.224	0.002	0.010
TIF	5.56	5.21	5.59	5.52	0.073	0.563	0.789
BIF	5.03	5.29	5.34	5.10	0.080	0.901	0.329
Excreta fat content	5.80	5.55	5.91	6.22	0.112	0.065	0.128

Values are mean of six pens of four birds each. Means in the same row without the same letter are significantly different ($P < 0.05$). AFR, abdominal fat rate, calculated as percentage of body weight; TIF, intramuscular fat content of thigh muscle; BIF, intramuscular fat content of breast muscle. AFR measurement was done as fresh basis; other measurements were done as dry basis.
[a] L, linear effect of increasing HFCP; Q, quadratic effect of increasing HFCP (0, 1, 3 and 5 g kg^{-1} diet).

supplementation with different levels of HFCP had no effect on intramuscular fat of thigh and breast muscles as well as the excreta fat content of broilers ($P > 0.05$). On both days 21 and 42 the abdominal fat rate decreased (L, $P < 0.01$; Q, $P < 0.05$) when the HFCP levels increased. A similar trend was observed for liver fat content (L, $P < 0.01$; Q, $P < 0.05$).

In recent years the high growth performance and improvements in properties of carcasses and meat quality have been beneficial to the poultry industry, especially in broiler production. In the meantime, consumers are becoming increasingly aware of the nutritional quality and health benefits of the food they consume.[31] In particular, the major emphasis has been on better body composition with higher breast meat yield and lower abdominal fat.[32] The present study showed that deposition of broiler abdominal fat could be reduced by supplementing HFCP in the diet. The lipid-lowering effect in broilers of other polysaccharides such as chitosan was also documented in earlier studies.[22,33] The hypolipidemic effects of HFCP demonstrated in the present study are not clearly understood. However, once probiotic bacteria colonize the gut, they ferment food-derived indigestible carbohydrates, leading to increased production of SCFA (e.g. acetic acid, propionic acid and butyric acid).[34] When SCFA are present in the gut, they are quickly absorbed by the large intestine and then metabolized by the liver.[14] Acetate is a lipogenic substrate whereas propionate inhibits lipogenesis in isolated hepatocytes of normal[35,36] and Zucker[37] rats. This action of propionate could involve a competition between propionate and acetate for the transporter of acetate into hepatocytes.[38,39] Delzenne *et al.*[40] reported that propionate, at concentrations found in the serum of the portal vein of oligofructose-fed rats (0.6 mmol L^{-1}), inhibited carrier-mediated acetate uptake in cultured isolated hepatocytes and was able to decrease fatty acid synthase (FAS) mRNA concentration in cultured hepatocytes and could thus constitute a putative mediator of the anti-lipogenic effect of dietary oligofructose. An increase in bacterial count or a change in the composition of the bacterial population in the large intestine would result in increased fermentation and SCFA production.[41] Depending on the concentration of each fatty acid produced, body fat deposition may thus be altered through this mechanism. The results of this study showed that dietary supplementation with HFCP firstly increased the caecum lactobacilli and bifidobacteria counts, then increased the propionic acid concentration in the caecum digesta and finally

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decreased the abdominal fat deposition of broilers. On day 21, abdominal fat had negative linear correlations with lactobacilli counts ($r = -0.999$) and propionic acid concentration ($r = -0.988$) in the caecum of broilers; on day 42, abdominal fat had negative linear correlations with bifidobacteria counts ($r = -0.984$) and propionic acid concentration ($r = -0.994$) in the caecum of broilers, confirming the above speculation.

CONCLUSION

HFCP increased lactobacilli and bifidobacteria counts, inhibited *E. coli* counts and increased propionic acid concentration in the caecum of birds and decreased abdominal fat deposition of the starter and finisher broilers. Therefore dietary supplementation with HFCP may lead to the development of low abdominal fat of broilers as demanded by health-conscious consumers.

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